

CHORDIUS

PIRATES-SEA-RAIL

Technical Architecture Documentation: A High-Performance Gaming
Framework combining LibGDX, Spring Boot, and Transactional Integrity.

01

System Foundations

| Technology Stack & Core Modules

- **Frontend Client:** Desktop-JVM built with LibGDX (Java 17).
Utilizes OpenGL-based rendering and Scene2D UI.
- **Backend Gateway:** Spring Boot 3 microservice acting as a secure proxy to the wallet system.
- **Wallet API:** Centralized Node.js service managing Jigsaw Coin balances and security audits.
- **Database:** Hibernate-managed PostgreSQL/MySQL for persistence.

| The Event Command System

The Command Pattern

Decouples overworld triggers from the render thread. Implements the EventCommand interface to queue complex cutscenes and scripted movements without blocking.

Lifecycle Methods

start() initializes the command, update(delta) handles per-frame animation, and isFinished() triggers the next command in the queue.

| Object Factory & Map Loading



Gate

Transition zones parsed from TMX map properties, handling target coordinates and map string references.



Chest

Interactive loot containers configured dynamically with currency amounts and item identifiers.



Enemy Event

Physical bounding boxes that trigger battle transitions based on enemy_group_id logic.

| Battle State Machine

The turn-based combat system executes via a deterministic State Machine:

- **TURN_START**: Sorting turnQueue by speed.
- **INPUT**: AI calculation or user UI selection.
- **EXECUTE_ACTION**: Resolving skills & reactions.
- **BATTLE_END**: Cleanup and wipeout checks.



| Elemental Reaction Mechanics

1.5

Reaction Multiplier

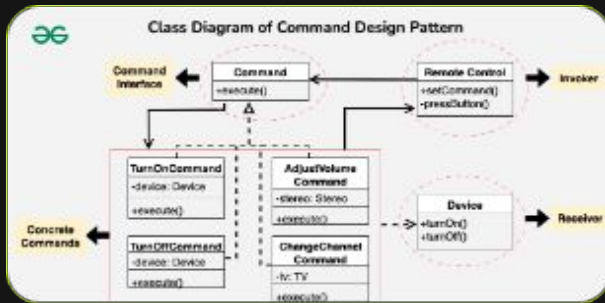
X

Dynamic Damage Resolution

Reactions like **Douse** and **Evaporate** trigger amplified damage based on dual-element marks. Formula for Reaction Damage:

$$DMG_{\text{Reaction}} = \frac{Lvl_1 + Lvl_2}{5} + 2 \times C_R \times \frac{MAT_1 + MAT_2}{MDF_T} \times 0.02 + 2$$

| Data-Driven Registry Architecture



Skill Database

Static attributes parsed from `skills.json` at startup, including costs and textures.



Skill Logic Registry

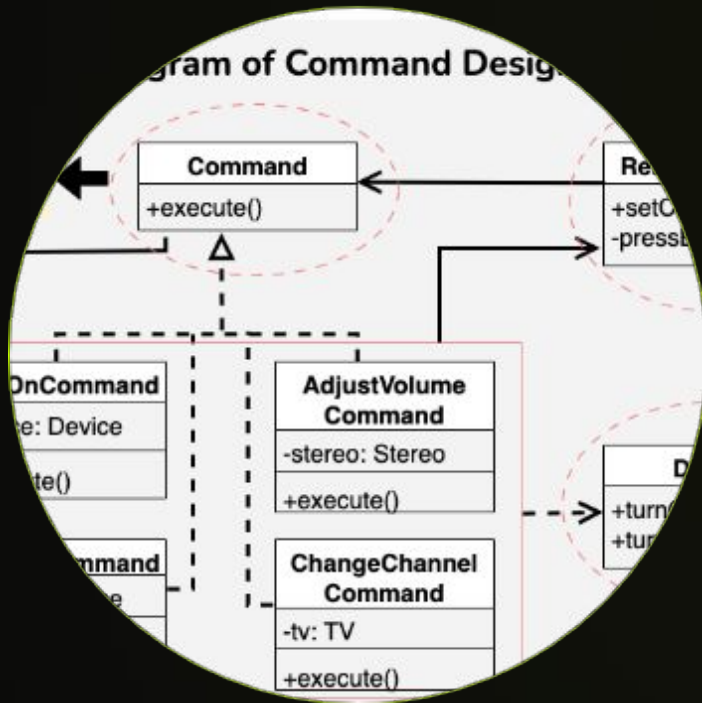
Maps string IDs to compiled Java logic implementations for complex card behaviors.



Script Registry

Routes Tiled map events to specific MapScript classes like `CherryTownScript`.

| Player Movement & State Pattern



Movement Delegation

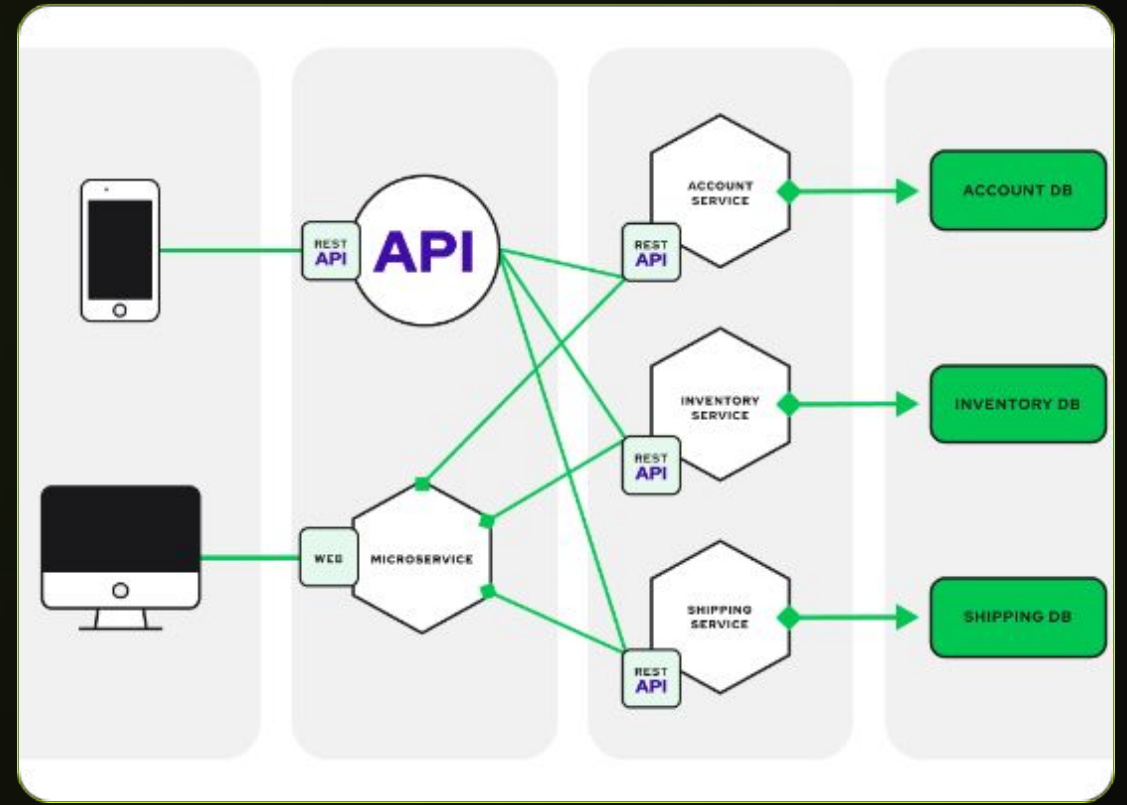
Physics and animations are decoupled via the **State Pattern**:

- **NormalState**: Standard WASD polling and physics.
- **DashingState**: Multiplies speed, spawns motion trails, and manages a cooldown timer.
- **Transition**: Fluid state switching prevents messy if-else branches in the player update loop.

| UI Observability & Listeners

Observer Pattern

To prevent tight coupling between Player and GameHUD, the HUD registers as a PlayerObserver. When health or status changes, the HUD updates reactively without frame-by-frame polling.



| Backend Transaction Workflow

Client Request

Auth or Jigsaw Coin
purchase initiated.

Node.js Wallet

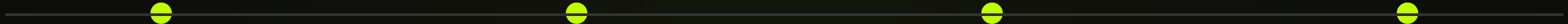
Global balance deduction
via secure API.

Spring Proxy

Validation and Order Log
creation.

DB Commit

Success response and
order fulfillment.

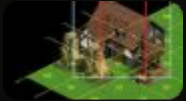


| Data Integrity & Schema

Table Entity	Identifier (PK)	Structural Purpose
Users	user_id (UUID)	Global profiles and currency totals.
Player Characters	instance_id	Character rosters, levels, and duplicates.
User Pity	pity_id	Persistent gacha pity counter per banner.
Local Orders	order_id	Audit trail for premium currency transactions.

**Character rosters in save.json are verified against the database on load to prevent save file manipulation.*

| Image Sources



<https://i.sstatic.net/OJVaS.png>

Source: stackoverflow.com



<https://i.redd.it/k3y9nlir8lil1.png>

Source: www.reddit.com



[https://media.geeksforgeeks.org/wp-content/uploads/20240202154433/CommandPatternExampledrawio-\(3\).png](https://media.geeksforgeeks.org/wp-content/uploads/20240202154433/CommandPatternExampledrawio-(3).png)

Source: www.geeksforgeeks.org



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Source: www.freepik.com



https://miro.medium.com/l*uxuLajvtDjLuhvw8xO2_Uw.png

Source: medium.com